

PROJECT DEFINITION

- We are here to develop an Adaptive AI-Based Educational Assessment Platform.
- A web-based Agentic AI assessment platform that hands in the users with :
 1. S1 Engine : Upload resources, such as PDF notes, from which the platform extracts relevant knowledge to generate customised quizzes, flashcards, topic-wise tests, and full-length assessments.
 2. S1 Engine : Upload resources, such as PDF notes, from which the platform extracts relevant knowledge to provide back users with either a video explanation backed by Google's Gemini-3.2 LLM or a text explanation of the content backed by OpenAI's ChatGPT-4 LLM.
 3. S2 Engine : This engine is specifically designed to assist users with academic examinations such as the SAT, ACT, GRE, GMAT, TOEFL, and similar exams. This is achieved by generating a virtual environment that simulates the actual exam.
- The platform leverages Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and adaptive assessment techniques to provide personalised and structured learning experiences.

Key Features :

1. Notes-Based Assessment Tool (Agentic AI S1)
2. Notes-Based Explainer Tool (Agentic AI S1)
3. Standardised Exam Simulation Tool (Agentic AI S2)
4. Retrieval-Augmented Generation (RAG) + Large Language Model (LLMs)
5. Adaptive Learning Engine
6. Interactive Assessment Modes (With custom difficulty scale)
7. Evaluation & Feedback system
8. Multi-Agent AI Architecture
9. User Memory & Learning History

What makes this product different from a generic chatbot :

A generic chatbot generates free-form responses based on user prompts. These responses are limited to the chat history and uploaded files provided within the chat itself.

This is the flow of a generic chatbot like OpenAI's ChatGPT or Google's NotebookLM :

User → Prompt → LLM → Response

This is how our system works :

User → Planner → Retriever → LLM → Validator → Test Engine → Evaluator → Feedback → Adaptation

The platform uses Retrieval-Augmented Generation to ensure that AI-generated content is grounded in user-provided notes and verified contextual information rather than relying solely on generalised LLM responses.

This agentic model differs from a generic chatbot in that it is a structured, task-oriented assessment system rather than a conversational AI. A typical chatbot simply generates free-form responses based on user prompts, often without context grounding, consistency, or measurable outcomes.

In contrast, this system operates through a defined pipeline that includes retrieval of user-specific content (via RAG), controlled question generation, evaluation of answers, and adaptive feedback. It produces structured outputs (like MCQs and scores), tracks performance over time, and adjusts difficulty based on user progress and choice.